

Final Project Report

Global Food Production Analysis (1961–2023)

1. Introduction

1.1 Project Overview

This project focuses on analyzing global food production data from 1961 to 2023 using Power BI.

The dashboard provides insights into production trends of major food commodities such as rice, wheat, maize, tea, coffee, fruits, and vegetables across different countries and years.

1.2 Objectives

Analyse global food production trends over time

Identify top food-producing countries

Compare crop-wise production patterns

Create an interactive Power BI dashboard for decision-making

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

Global food demand is increasing rapidly. Understanding historical food production trends helps governments and organizations plan food security and agricultural policies effectively.

2.2 Project Proposal (Proposed Solution)

To develop an interactive Power BI dashboard that visualizes global food production data by country, year, and crop type, enabling users to gain meaningful insights easily.

2.3 Initial Project Planning

Understanding project requirements

Identifying datasets

Data cleaning and preprocessing

Dashboard design and development

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

The data was collected from publicly available global agriculture datasets containing yearly food production statistics for various countries.

3.2 Data Quality Report

Checked for missing values

Removed duplicates

Standardized country and crop names

Ensured numerical consistency

3.3 Data Exploration and Preprocessing

Power BI Power Query was used to:

Filter unnecessary columns

Aggregate yearly data

Create calculated measures

4. Data Visualization

4.1 Framing Business Questions

Which countries are the highest food producers?

How has food production changed from 1961 to 2023?

Which crops dominate global production?

4.2 Developing Visualizations

The following visuals were created:

KPI Cards

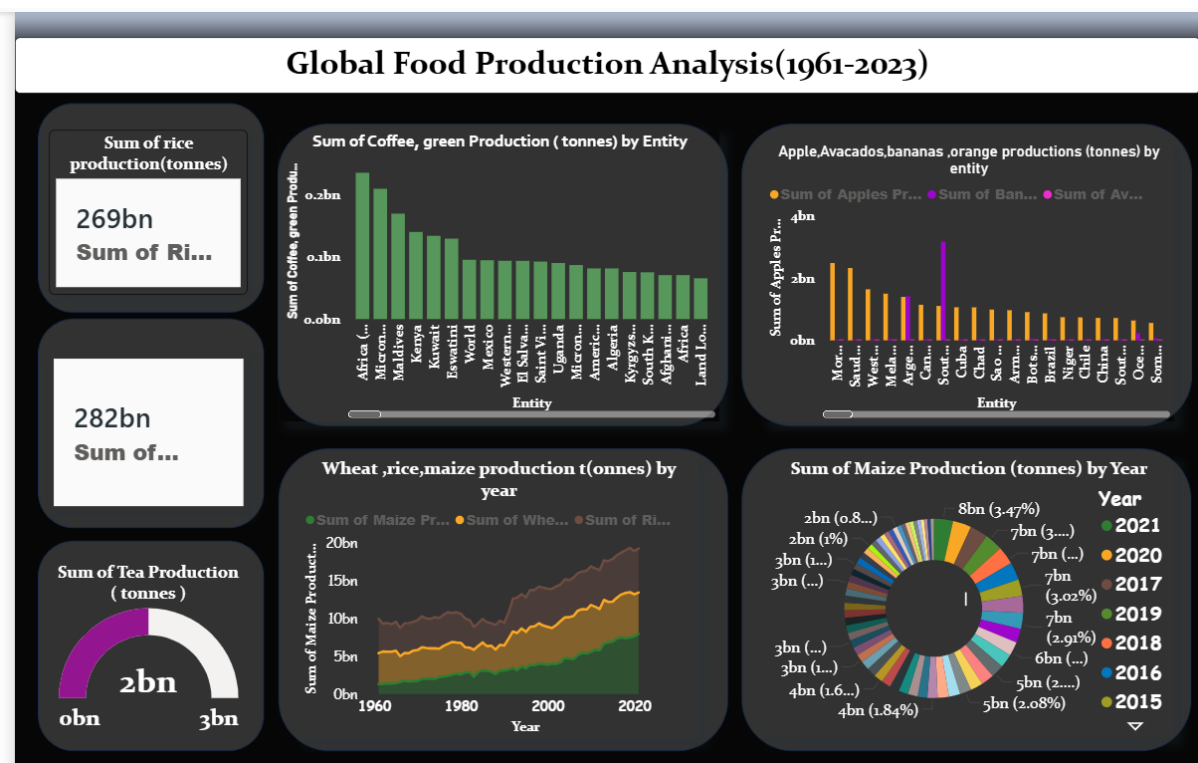
Bar Charts

Line Charts

Donut Charts

Trend Analysis Graphs

5. Dashboard



5.1 Dashboard Design File

The dashboard is designed with a dark theme and includes:

Total production KPIs

Country-wise comparison

Crop-wise production analysis

Year-wise trend analysis

6. Report

6.1 Story Design File

The report presents insights in a structured and visually appealing manner, enabling users to explore food production data interactively.

7. Performance Testing

7.1 Utilization of Data Filters

Filters applied:

Year

Country

Crop Type

7.2 Number of Calculated Fields

Multiple DAX measures were created, including:

Total Production

Year-wise Aggregates

7.3 Number of Visualizations

The dashboard contains 10+ interactive visualizations.

8. Conclusion / Observation

The analysis shows a significant increase in global food production over the decades, with certain countries consistently leading in production. Crop-wise trends reveal changing consumption and agricultural focus.

9. Future Scope

Predictive analysis using machine learning

Real-time data integration

Climate impact analysis on food production

10. Appendix

10.1 Source Code

Power BI DAX measures and Power Query transformations were used.

10.2 GitHub & Project Demo Link

GitHub repository and Power BI Service dashboard link